

Regret Is Weighted Forgetting

Electronic supplementary material

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This electronic supplementary material contains content moved from the main article to meet the Proceedings A page limit.

Table S1: Logical dependencies. Each result can be verified by checking only the items listed. Appendix entries are imported results, all proved in Section S10.

Result	Depends on
Theorem 3.10 (exact reduction)	Definitions 3.1 to 3.4, Lemma 3.7, and Definition 3.8 (self-contained)
Corollary 3.14 (policy-wise decomposition)	Theorem 3.10
Corollary 3.15 (representation search)	Theorem 3.10
Theorem 3.18 (multi-class)	Self-contained (the cellwise argument of Theorem 3.10)
Proposition 5.1 (regret preorder)	Theorem 3.10
Theorem 5.2 (diagnostic-quotient convergence)	Theorem 3.10
Proposition 5.3 (Lipschitz stability)	Definition 3.4 and Theorem 3.10
Corollary 4.10 (Stack-Theoretic form)	Theorem 3.10, Definition 4.7, and Proposition 4.9
Proposition 4.11 (freedom deficit dominates Bayes error)	Theorem 3.10 and the setting of Corollary 4.10
Corollary 5.4 (generalisation probability)	Corollary 4.10 and Proposition 4.9

S1 Recurrent Proximal Policy Optimization (PPO) representation search

This section gives full methods and checks for the experiment in the main article. A *world* is one sampled delayed-cue environment with fixed binary test mappings, cue reliabilities, and mixture frequency for two episode conditions. A *fixed cohort* evaluates worlds without conditioning on selector disagreement, so agreements remain. Each member is a *fixed world*. A *screen cohort* retains worlds only when selectors disagree under pre-target screening. I call each retained case a *stable-disagreement world*. A *decision margin* is the normalised value lost by choosing the worse

binary answer. I use *target truth* for hidden target classes, posterior class probabilities, correct answers, rewards, and regret. I use *KBT* as a short label, not an acronym, for the target-input K_ρ selector, which uses source-estimated margins and target histories without target truth. *Raw impurity* uses the same target histories and source-estimated answers after setting every decision margin to one. A *leave-one-panel-fold-out fit* recomputes a selector after omitting one of eight disjoint panel folds. A choice is *fold-stable* when the selected candidate remains unchanged across all eight fits. A *rank-one deployment* deploys the candidate ranked first by a selector. A *cell policy* assigns one binary answer to each representation-test cell. A *source stage* is all training, calibration, admission, and selection completed without target truth. It is *locked* once candidate, cell-policy, and hash records are written before target truth is opened. I call these records the *lock*. The frozen plan defines four cohorts. Fixed A and fixed B each contain 24 fresh worlds and retain selector agreements. Screen A and screen B examine fresh worlds until each has 24 stable-disagreement worlds. Target truth remains sealed until all four source stages have locked. Screen A retained 24 worlds from 128 screened worlds, while screen B retained 24 from 96.

S1.1 Worlds and shifted episode sets

Each world has four hidden classes, three binary tests, three cue events, and five distractor symbols. Hidden class g is uniform on $\{1, 2, 3, 4\}$. Each binary test j maps two classes to answer 0 and two to answer 1. The three test-answer vectors distinguish all four hidden classes. Test mappings stay fixed between source and target.

Each episode carries a visible flag indicating one of two cue-reliability and cue-placement conditions, which I call *regimes*. Rare-regime probability is sampled once per world from $\text{Unif}(0.08, 0.16)$. Rare episodes use cue reliability $q_H \sim \text{Unif}(0.87, 0.96)$ and place three cues immediately after episode start. Common episodes use $q_L \sim \text{Unif}(0.27, 0.38)$ and place three cues immediately before the first test. Conditional on hidden class g , a cue equals g with probability q_r in regime r and is otherwise uniform over the three other classes. Tests occur after delay d , then three and six steps later. Source delays are $d \in \{3, 5, 7\}$. Target delays are $d \in \{12, 20, 28\}$. Cue channel, regime mixture, binary test mapping, and reward stay fixed. Only delay distribution changes.

Answers receive unit reward when correct at a test and zero otherwise. Answers do not affect later observations. This design isolates delayed memory and representation aliasing. It does not test exploration or action-dependent transition control.

A *quantiser* maps a recurrent hidden state to one of six categorical memory states. I use *bank* for an episode set reserved for one protocol role. Each world has a source-fit bank of 3,000 episodes for fitting quantisers, a separate source-calibration bank of 1,000 labelled episodes for cue-reliability estimation, and a source-validation bank of 2,400 episodes for candidate admission. A *target panel* is an input-only bank of 1,600 target episodes used for selection. A *fresh target bank* is a separate 2,400-episode target set used only after every source stage is locked. The target panel contains observations, valid-step masks, test locations, visible regime flags, cue symbols, delays, lengths, and a seed. It contains no hidden class, correct answer, posterior, reward, or regret.

For cues c_{i1}, c_{i2}, c_{i3} in regime r_i , the hidden-class posterior is

$$p_i(g) \propto \prod_{\ell=1}^3 \begin{cases} q_{r_i}, & c_{i\ell} = g, \\ (1 - q_{r_i})/3, & c_{i\ell} \neq g. \end{cases} \quad (\text{S1})$$

Let $\tau_j(g) \in \{0, 1\}$ be the correct answer for class g on test j . Expected value of binary answer b is

$$V_{ij}(b) = \sum_{g=1}^4 p_i(g) \mathbf{1}\{\tau_j(g) = b\}. \quad (\text{S2})$$

The selector uses the same equations with source estimates $\hat{q}_H$ and $\hat{q}_L$. Define

$$\hat{y}_{ij} = \arg \max_{b \in \{0,1\}} \hat{V}_{ij}(b), \quad \hat{\rho}_{ij} = 1 - \frac{\min_b \hat{V}_{ij}(b)}{\max_b \hat{V}_{ij}(b)}. \quad (\text{S3})$$

Thus $\hat{\rho}_{ij}$ is the estimated normalised loss from choosing the worse binary answer. No target outcome is needed to compute it on the input-only panel.

S1.2 PPO training and candidate representations

Each world trains six source-only recurrent Proximal Policy Optimization (PPO) agents [21]. Encoders are either gated recurrent unit (GRU) or long short-term memory (LSTM) networks with 32 hidden units. For each architecture, PPO episodes use the natural regime frequency, rare-regime frequency 0.50, or rare-regime frequency 0.02. Optimisation runs for 120 updates with 192 episodes per update, two epochs, minibatches of 192, learning rate 0.0025, entropy coefficient 0.01, discount 0.99, Generalized Advantage Estimation (GAE) coefficient 0.95, PPO clip 0.2, value coefficient 0.5, and gradient-norm limit 1.0. Snapshots at updates 60 and 120 yield 12 encoders per world.

For each episode, one 32-dimensional hidden state is read immediately before the first test and reused as memory for all three tests. Quantisation begins with one *affine row frame*, a coordinate system fitted from the encoder’s source-fit hidden rows. The first source row is the origin. A deterministic full-rank set of source rows defines an affine basis. Every source, panel, and target hidden row is expressed in this basis, rounded to ten decimal places, and standardised in the resulting source coordinates. The same frozen source frame is reused across all five memory allocations for that encoder.

K-means is fitted separately within rare and common regimes. Five allocations divide six categorical states between regimes as (5, 1), (4, 2), (3, 3), (2, 4), and (1, 5). This gives 60 candidate representations per world. Every candidate therefore has six states and at most 18 representation-test cells. Candidate differences come from encoder architecture, training emphasis, checkpoint, and state allocation rather than memory capacity.

The affine row construction is invariant to dense invertible affine recoding and affine shifts of the recurrent hidden coordinates. Quantiser receipts record the origin, basis rows, basis, coordinate normalisation, centroids, frame rule, and a cryptographic hash computed with the Secure Hash Algorithm using a 256-bit digest (SHA-256). The audit applies dense recodings, shifts, and within-regime category relabellings and verifies unchanged scores and selected identities. All recorded encoders had full affine rank 32.

Let $R_s(M)$ be empirical mean reward on the independent source-validation bank under a source-fitted cell policy. A candidate is admitted when $R_s(M)$ lies within 0.025 of the best candidate in the same world. If fewer than 12 pass, the top 12 are admitted.

S1.3 Selectors, screening, and lock

Given target-panel codes $M(h_i)$ and source-estimated binary terms, KBT score is

$$\hat{K}_{\rho,T}(M) = \frac{1}{3N} \sum_{j=1}^3 \sum_c \min_{b \in \{0,1\}} \sum_{i: M(h_i)=c} \hat{\rho}_{ij} \mathbf{1}\{\hat{y}_{ij} \neq b\}. \quad (\text{S4})$$

The fitted policy chooses the minimising answer in each cell. This is literal binary K_ρ from Theorem 3.10, evaluated with source-estimated margins on unlabelled target histories.

Raw impurity uses the same target histories and hard pseudo-answers $\hat{y}_{ij}$,

$$I_T(M) = \frac{1}{3N} \sum_{j,c} \min_{b \in \{0,1\}} \sum_{i: M(h_i)=c} \mathbf{1}\{\hat{y}_{ij} \neq b\}. \quad (\text{S5})$$

It discards answer-value margins. The primary comparator first selects a representation by $I_T(M)$ and then fits the same margin-weighted target-panel policy used after KBT selection. This isolates representation selection. Source- K_ρ minimises a source-bank score. Source validation maximises source-validation reward. Uniform random is the admitted-pool mean. Target oracle minimises fresh target K_ρ after every source stage is locked.

Fixed A and B retain all worlds, including those where KBT and raw impurity choose the same representation. Their primary endpoint compares, within each world, Spearman correlation between selector score and fresh target K_ρ over admitted candidates. Rank-one fresh target K_ρ is a deployment endpoint.

Stable-disagreement screening uses no fresh target outcome. The 1,600-history target panel is divided into eight folds. KBT and raw impurity are each recomputed after leaving out one fold at a time. A screened world is retained only when pooled selectors differ and each pooled identity remains unchanged in all eight fits. Sixteen source-calibration folds are recorded as an audit of calibration stability and do not alter screening. Each screening replication stops after 24 retained worlds or at a frozen cap of 160 worlds.

Before target truth is opened, every source world writes its source banks, input-only target-panel hash, cue estimates, admitted candidate identifiers, selector identifiers, panel-fold decisions, source-calibration-fold audit, cell policies, quantiser receipts, and SHA-256 lock record. The driver writes one opening record only after all fixed and screening source stages finish. Fresh target evaluation then checks all hashes before computing hidden-class posteriors, correct answers, K_ρ , regret, reward, and success. Nonselected screening worlds receive no target evaluation.

S1.4 Outcomes and statistical analysis

Fresh target K_ρ is computed from target truth on the new 2,400-episode bank. For action a_{ij} chosen by a locked cell policy, target normalised regret is

$$\bar{\delta} = \frac{1}{3N} \sum_{i,j} \left(1 - \frac{V_{ij}(a_{ij})}{\max_b V_{ij}(b)} \right). \quad (\text{S6})$$

Target success is the fraction of answers matching $\tau_j(g_i)$ for sampled hidden classes. Expected target reward uses posterior answer value. Native PPO target return evaluates the selected recurrent actor without the fitted cell policy.

Fixed-world primary result is the paired difference in candidate-ranking Spearman correlation. Fixed deployment result is $K_\rho(M_{\text{raw}}) - K_\rho(M_{\text{KBT}})$ with selector agreements retained. Stable-disagreement primary result is the same target K_ρ difference conditioned on panel-fold-stable selector disagreement. Positive values favour KBT. Means are world means. Intervals are percentile bootstrap intervals over worlds with 50,000 resamples. One-sided sign-flip values are exact for 24-world replications and use one million fixed-seed draws for 48-world combinations. The two stable-disagreement replication values are also combined by Fisher’s method.

Table S2: Fixed-world and stable-disagreement results. Ranking gain is KBT Spearman correlation minus raw-impurity Spearman correlation. Cost gains are raw impurity minus KBT. Success gain is KBT minus raw impurity.

Comparison	KBT	Raw impurity	Paired gain	95% interval
Fixed candidate-ranking correlation	0.959470	0.910559	0.048911	[0.032299, 0.066819]
Fixed target K_ρ	0.100015	0.101843	0.001828	[0.000748, 0.003016]
Fixed target regret	0.101469	0.104567	0.003098	[0.001795, 0.004601]
Fixed target success	0.545249	0.539300	0.005949	[0.003872, 0.008163]
Stable-disagreement target K_ρ	0.090744	0.096840	0.006096	[0.004713, 0.007485]
Stable-disagreement target regret	0.091498	0.099838	0.008340	[0.006753, 0.009916]
Stable-disagreement target success	0.541670	0.528626	0.013044	[0.010697, 0.015593]

Fixed-world ranking gain was positive in 44 of 48 worlds, with four losses and $p < 10^{-6}$. Rank-one fixed target K_ρ had 20 wins, 25 agreements, and three losses, with $p = 0.000720$. The same direction held for fixed target regret and success.

Stable-disagreement target K_ρ fell by 6.3% relative to raw impurity. KBT won 43 of 48 worlds and lost five. Replication A gain was 0.006581 with interval [0.004870, 0.008294] and exact $p = 4.77 \times 10^{-7}$. Replication B gain was 0.005610 with interval [0.003498, 0.007809] and exact $p = 1.91 \times 10^{-5}$. Fisher-combined p was 2.40×10^{-10} . Stable-disagreement regret fell by 8.4%. Success increased in 47 worlds and tied in one.

Table S3: Mean target normalised regret by delay over 48 stable-disagreement worlds.

Delay	KBT	Raw impurity	Random mean	Target oracle
12	0.092269	0.099111	0.115088	0.089697
20	0.089358	0.099315	0.116097	0.088441
28	0.092785	0.101080	0.117897	0.091969

S1.5 Estimation and audit

Across all 2,880 fixed-world candidates, estimated and fresh target K_ρ have mean within-world Spearman correlation 0.962606. Mean absolute error is 0.006127, median 0.004987, and maximum 0.031374. KBT matches the target-oracle representation in 39 of 48 fixed worlds. Mean target-oracle K_ρ gap is 0.000589. Median cell occupancy is 18 of 18. Exactly 2,686 of 2,880 candidates occupy all 18 possible representation-test cells.

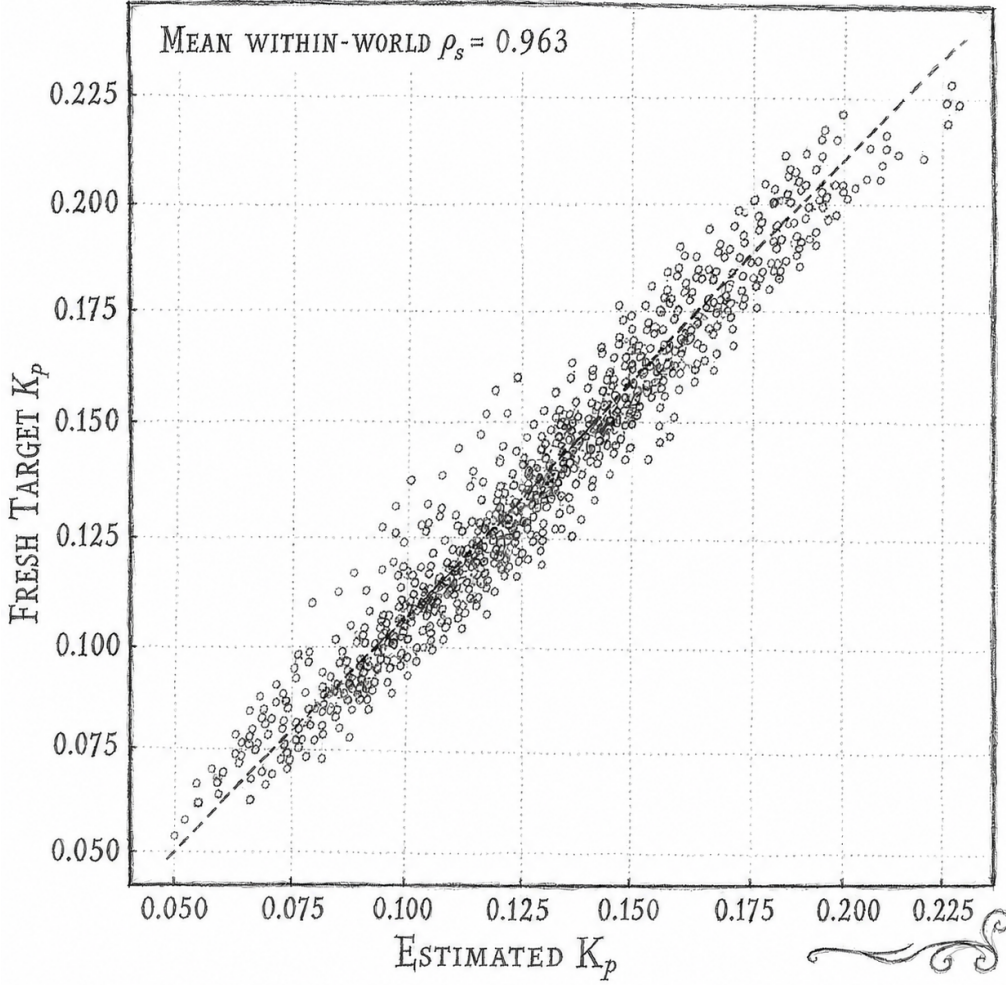


Figure S1: Source-calibrated target-panel estimate against fresh target K_ρ for all 2,880 candidates in 48 fixed worlds. Dashed line is equality.

The compact experiment archive includes the frozen plan, source code, selected-world candidate tables, selector locks, quantiser receipts, screening records, cohort summaries, figures, statistical recomputation, and independent audits. The recorded result passes 30 result checks. Fresh statistical recomputation reproduces fixed ranking gain 0.048910982460, fixed target K_ρ gain 0.001828109631, and stable-disagreement target K_ρ gain 0.006095510809. The compact archive omits deterministic PyTorch states and generated input arrays, which can be regenerated from frozen seeds and settings.

Audits independently recompute source and target score formulas, selected identities, fitted policies, target results, affine geometry, dense affine recoding invariance, affine-shift invariance, within-regime category relabelling, panel-fold screening, source-calibration folds, lock-record hashes, target-panel hashes, and source-before-target chronology. Every recorded cohort and screening audit passes.

S1.6 Scope

This study tests source-calibrated target-input representation selection under delayed partial observability and covariate shift. It does not introduce a general-purpose reinforcement learning (RL)

algorithm. Target cue channel and binary test mappings remain invariant while delay changes. Answers do not affect later observations. Six-state categorical representations are quotients of 32-unit recurrent states rather than recurrent networks trained with six-unit memory. Cellwise policies are fitted after PPO encoder training, so the experiment isolates representation selection. Fixed worlds estimate unconditional deployment benefit. Stable-disagreement worlds estimate selector quality conditional on robust pre-target disagreement. Action-dependent control, broader memory benchmarks, and end-to-end optimisation of a differentiable estimator remain open.

S2 Earlier diagnostic experiments

The theorems above are exact algebraic identities, so they do not require empirical validation. The purpose of this section is to show that $K_\rho(M)$ is informative in practice. It shows that the bridge identity holds under native Stack-Theoretic computation, that K_ρ discriminates between representations where simpler diagnostics do not, and that the aliasing-execution decomposition tracks representation quality during training. The two-sided verification below cross-checks the Stack-Theoretic reformulation of Section 4.5 against the RL computation. The remaining studies evaluate K_ρ purely as an RL diagnostic and use none of the bridge machinery.

S2.1 Two-sided verification on Boolean domain

Here the diagnostic task is encoded natively using the Stack Theory Suite (STS) code library [6]. The environment has $n_{\text{bits}} = 5$ Boolean variables, giving $2^5 = 32$ states. The vocabulary contains 10 programs (bit-tests $b_i=0$ and $b_i=1$ for each bit i), giving an induced language $|L_v| = 243$. For each trial, a random structured Boolean full-state label $Y : \{0, 1\}^5 \rightarrow \{0, 1\}$ is generated that depends on 2–3 randomly chosen bits. Representations are defined by which bits the agent observes (0 through 5). The diagnostic support points are partial observation histories rather than necessarily full Boolean states. For a partial observation pattern s , the diagnostic probability $p(s)$ is the posterior probability that $Y = 1$ after marginalising over full Boolean states consistent with s . Thus full-state labels are deterministic, while diagnostic probabilities can still be near $1/2$.

The bridge is verified two-sided through independent code paths. On side A, the RL side, cells are computed by direct bit inspection and K_ρ is computed via the cellwise formula $\sum_C \min\{A_C, B_C\}$. On side B, the Stack-Theoretic side, the STS statement for each possible observation pattern is built by conjoining the appropriate bit-test programs. Its truth set is computed via `Statement.truth_set()`. The states in the truth set define the cell membership. K_ρ is then recomputed from the STS-derived cells. Both the cell structures and the resulting K_ρ values are compared. Over $100 \text{ trials} \times 6 \text{ representation levels} = 600$ checks, cell membership agrees in every case, and K_ρ matches to numerical precision. The two-sided agreement confirms that the Stack-Theoretic reformulation is not a post-hoc relabelling. The STS truth-set computation and the RL cellwise computation produce identical cell structures through entirely independent formal machinery.

Table S4 shows that K_ρ and raw impurity behave differently. K_ρ reaches zero at 4 observed bits. The remaining unobserved bit creates cells that are impure but whose impurity bears zero margin weight (the minority class within each cell has $\rho_i \approx 0$ because the posterior is close to a coin flip on that bit). Raw impurity remains positive until all 5 bits are observed. In a separate comparison of 200 pairs of 2-bit versus 3-bit representations, K_ρ discriminated between representations that had similar accuracy ($< 5\%$ difference) in 21% of cases.

Table S4: Boolean classification experiment (5 bits, 32 states, 100 trials). K_ρ is computed via RL cellwise arithmetic (side A) and independently via STS truth-set enumeration (side B). Both agree to numerical precision in all 600 checks. Cell cardinality is deterministic given the number of observed bits.

Bits observed	$K_\rho(M)$	Raw impurity	Accuracy	Cells
0	0.102 ± 0.008	0.318 ± 0.015	0.68 ± 0.01	1
1	0.079 ± 0.007	0.285 ± 0.014	0.71 ± 0.01	2
2	0.047 ± 0.007	0.247 ± 0.015	0.75 ± 0.02	4
3	0.019 ± 0.005	0.191 ± 0.015	0.81 ± 0.01	8
4	0.000 ± 0.000	0.124 ± 0.016	0.88 ± 0.02	16
5	0.000 ± 0.000	0.000 ± 0.000	1.00 ± 0.00	32

Table S5: Discretised MLP encoder on a 16-state POMDP (20 trials). Finer discretisation reduces both K_ρ and raw impurity, but K_ρ drops faster. At 6 bins, K_ρ is a factor of $5.5\times$ smaller than raw impurity.

Bins	$K_\rho(M)$	Raw impurity	Accuracy	Cells
2	0.0097 ± 0.0004	0.041 ± 0.001	0.959 ± 0.001	166
3	0.0101 ± 0.0008	0.038 ± 0.002	0.962 ± 0.002	172
4	0.0057 ± 0.0004	0.026 ± 0.001	0.974 ± 0.001	343
5	0.0036 ± 0.0003	0.017 ± 0.001	0.983 ± 0.001	378
6	0.0022 ± 0.0002	0.012 ± 0.001	0.989 ± 0.001	517
8	0.0009 ± 0.0002	0.005 ± 0.001	0.995 ± 0.001	630
10	0.0003 ± 0.0000	0.002 ± 0.000	0.998 ± 0.000	700
15	0.0001 ± 0.0000	0.000 ± 0.000	1.000 ± 0.000	763

S2.2 Discretised encoder at varying granularity

To demonstrate the diagnostic value of K_ρ at moderate scale, I construct a structured partially observable Markov decision process (POMDP) with $|\mathcal{S}| = 16$ hidden states, $|\mathcal{O}| = 6$ observations, and observation sequences of length $L = 8$. The first 8 states have label $y = 1$ and the remaining 8 have label $y = 0$. Observation distributions are block-structured so that states in the same class share similar emission profiles, with cross-block transitions occurring infrequently.

I train a multilayer perceptron (MLP) with two layers, hidden dimension 4, and tanh activation to convergence on each of 20 random POMDPs, then evaluate the learned representation at 8 levels of discretisation granularity (from 2 to 15 bins per hidden dimension). Finer discretisation yields more cells and a more refined partition of the observation-history space.

Table S5 shows that K_ρ is consistently smaller than raw impurity across all granularities, because it discounts aliasing on near-tie diagnostic tests. At 6 bins per dimension, $K_\rho = 0.002$ while raw impurity is still 0.012, a $5.5\times$ ratio. Both quantities eventually approach zero at fine granularity, but the intermediate gap holds the most diagnostic information. A representation that looks impure by raw vote may have near-zero aliasing cost on tests that affect decision-making.

S2.3 Representation search across architectures

To test whether K_ρ contains actionable information beyond accuracy, I compare four representation architectures on the same 16-state structured POMDP. The four are a GRU (hidden dimension 4, trained for 300 epochs on sequences), an MLP (hidden dimension 4, trained for 400 epochs on

Table S6: Architecture comparison on a 16-state POMDP (20 trials). The GRU has the lowest K_ρ but the MLP has the highest accuracy. In 35% of trials, the architecture with the best K_ρ differs from the one with the best accuracy.

Architecture	$K_\rho(M)$	Accuracy	Raw impurity	Cells
GRU	0.0031 ± 0.0004	0.978 ± 0.002	0.022 ± 0.002	50
MLP	0.0039 ± 0.0004	0.982 ± 0.001	0.018 ± 0.001	371
Last-obs	0.0683 ± 0.0026	0.838 ± 0.004	0.162 ± 0.004	6
Random	0.1679 ± 0.0057	0.755 ± 0.006	0.245 ± 0.006	212

Table S7: Decomposition for a learned MLP during training on the 16-state POMDP ($L = 8$, hidden dim 4, 20 trials). The ratio $K_\rho/(K_\rho + \text{exec. cost})$ drops from 0.30 to 0.03 in the first 100 epochs, making the transition from representational bottleneck to policy bottleneck visible.

Epoch	$K_\rho(M_\theta)$	Execution cost	K_ρ / total	Accuracy
0	0.112 ± 0.005	0.268 ± 0.005	0.30	0.49 ± 0.01
50	0.012 ± 0.001	0.158 ± 0.005	0.07	0.81 ± 0.00
100	0.003 ± 0.000	0.097 ± 0.002	0.03	0.83 ± 0.00
150	0.003 ± 0.000	0.090 ± 0.001	0.03	0.83 ± 0.00
250	0.003 ± 0.000	0.086 ± 0.001	0.03	0.83 ± 0.00
500	0.003 ± 0.000	0.085 ± 0.001	0.04	0.83 ± 0.00

flattened one-hot inputs), a last-observation-only baseline, and a random projection (tanh of a fixed random linear map). All representations are discretised with 5 bins per hidden dimension.

Table S6 reveals that the GRU achieves the lowest irreducible regret ($K_\rho = 0.003$), while the MLP achieves the highest accuracy (0.982). The MLP achieves higher end-to-end accuracy even though its discretised representation has slightly higher irreducible aliasing cost. This shows that accuracy includes effects of the trained classifier and the continuous hidden geometry available to that classifier, whereas K_ρ isolates the aliasing cost of the chosen representation-test quotient after discretisation. In 35% of trials, the architecture with the best K_ρ differs from the one with the best accuracy. This confirms that K_ρ captures a distinct aspect of representation quality, namely the structural aliasing cost, that accuracy conflates with policy optimisation. For the finite candidate family in this experiment, Corollary 3.15 selects the GRU by minimum K_ρ , while end-to-end accuracy selects the MLP. This is a representation search over a declared family. It shows that the two criteria rank candidates differently, rather than that either criterion dominates for every downstream objective.

S2.4 Learned representation during training

To demonstrate the decomposition as a training diagnostic, I train a two-layer MLP classifier (hidden dimension 4, tanh activation, discretised to 5 bins per hidden unit) on the 16-state structured POMDP with $|\mathcal{O}| = 6$ observations and observation sequences of length $L = 8$. The discretised hidden-layer activations define the representation-test cells.

Table S7 reports $K_\rho(M_\theta)$, execution cost, and their ratio at selected training epochs, averaged over 20 trials.

K_ρ drops by a factor of $37\times$ in the first 100 epochs (from 0.112 to 0.003) as the hidden layer learns to separate the two state classes. After that, K_ρ plateaus near zero and further improvements come from execution cost reduction. The ratio $K_\rho/(K_\rho + \text{exec. cost})$ falls from 0.30 to 0.03 during this

period, making the bottleneck transition quantitatively visible. Early in training, the representation is the bottleneck. Later, the policy is the bottleneck. The decomposition renders that transition readable from a single diagnostic, which is the empirical contribution of the bridge for representation evaluation.

S3 Additional related work

S3.1 Predictive state and world models

A parallel line of work asks what internal structure low-regret agents must have by reasoning about predictive sufficiency. Littman, Sutton, and Singh introduced predictive state representations (PSRs), showing that observable predictions can serve as a complete state description without latent-variable modelling [16]. Singh, James, and Rudary extended the framework to controlled systems [22], and Boots, Siddiqi, and Gordon connected PSR learning and planning [9]. The approximate information state literature addresses partial observability more directly by identifying sufficient statistics for near-optimal planning [23].

On the model-learning side, Ha and Schmidhuber’s world models [12], Hafner et al.’s Dreamer agent [13], and Schrittwieser et al.’s MuZero [20] demonstrate that learned latent dynamics can support competitive planning. Richens, Everitt, and Abel give a theoretical counterpart, proving that general agents need world models under distributional shift [19]. My bridge theorem gives a different angle on the same family of questions. Rather than asking what structure a model must have, it asks how much regret a fixed representation forces, measured in weighted forgetting terms.

S4 Additional bridge context

S4.1 Enactive General Reinforcement Learning (EGRL) and neural completion measures

EGRL is the Stack-Theoretic translation from reward-labelled interaction histories into the task-extension language. A pair proxy is the rule used to rank the admissible positive-negative policy pairs extracted from that history. The EGRL freedom pair proxy ranks such pairs by the size of their joint extension. A separate neural study asks whether sampled joint label commitments remain feasible under one shared affine head on a frozen representation [7]. I treat that study as external evidence rather than part of the present theorem. Prior work constructs instantiated history tasks, reward predicates, admissible pair policies, selective memory for inconsistent histories, and EGRL-style translations from interaction histories into the task-extension language [3, 5, 6, 8]. The present theorem forms a mathematical core for that broader programme. It fixes a diagnostic goal family, sorts the history support into representation-test cells, and proves that the resulting irreducible error is weighted selective forgetting.

Interpretation. The earlier programme translated the whole interaction loop. This paper isolates the reusable kernel, which is the irreducible error at the representation-test cells.

S5 Representation search and external evidence

The bridge identifies $K_\rho(M)$ as the regret that no policy reading representation M can remove. An independent neural study evaluates joint completion on frozen representations [7]. The two

quantities answer different questions. $K_\rho(M)$ measures ambiguity forced by a representation-test quotient. The neural joint-bank score measures how often sampled multi-point class commitments remain feasible under one replacement affine head.

S5.1 Representation search in reinforcement learning

Corollary 3.15 turns K_ρ from a fixed-representation diagnostic into a representation-selection criterion. Given a declared family $\mathcal{M}$, one first selects a representation minimising $K_\rho(M)$ and then selects a cellwise optimal policy for that representation. The family can restrict memory states, encoder architecture, discretisation, latency, or another embodied resource.

Earlier discretisation and architecture studies in Section S2 are finite diagnostic instances of this search. Recurrent PPO study in Section 6 applies criterion to learned neural states. Source-labelled calibration estimates cue conditionals and margins, while unlabelled target histories identify target quotient cells. Proposition 5.3 bounds score perturbation from margin estimation error. This supports post-training selection among candidate encoders and quantisers. It does not yet optimise recurrent representation end to end.

S5.2 Candidate representation losses

The identity still suggests auxiliary losses for recurrent encoders on partially observable tasks. Given a hidden-state predictor above the encoder, one can weight prediction loss by the decision margin ρ_i , concentrating training on distinctions that affect the diagnostic decision. One can instead use a description-length penalty to remove irrelevant variation from spare representational capacity. These losses act on the representation. Margin weighting is a surrogate for reducing the diagnostic aliasing cost. Description-length regularisation is a comparator. Neither computes a policy extension.

Preliminary experiments on a binary-context corridor, T-Maze, NoisyCartPole with hidden velocities, and a symbol-recall task show task-dependent rankings. Margin weighting helps when the diagnostic target and control target coincide. Description-length regularisation helps when the encoder has excess capacity relative to the task. No tested surrogate dominates across all four tasks. The result is consistent with the distinction between representation loss and policy freedom.

S5.3 External evidence for freedom

A separate study operationalises freedom on frozen neural representations [7]. It fixes child labels and anchor predictions, then tests a predeclared bank of 512 joint probe-label bundles. One replacement affine head must satisfy each bundle together with the empirical policy. The joint bank score J_H is the fraction of feasible bundles and is invariant under invertible affine changes of feature coordinates.

Across predeclared cohorts of 100 networks trained on the Modified National Institute of Standards and Technology (MNIST) handwritten-digit benchmark and 100 networks trained on the Fashion-MNIST clothing-image benchmark, J_H has Spearman correlations 0.290 and 0.468 with held-out accuracy. Raw Hessian trace and final-layer relative flatness have no multiplicity-corrected association. Spectral-product margin has larger point estimates, and smaller weight norm also predicts accuracy. A matched random-label cohort has correlation 0.058 despite greater mean bank feasibility. The paper therefore reads J_H as task-specific external-demand survival rather than a universal score across unrelated tasks [7].

I use this evidence without importing the paper’s formal construction, common-task theorem, probably approximately correct Bayesian certificates (PAC-Bayes certificates), rescaling analysis, or experiments into this paper. Those results remain claims of the cited paper.

Earlier inductive logic programming experiments compared freedom selection with minimum description length on binary addition and multiplication. Freedom selection achieved 1.1–5 times the generalisation rate [1]. Those experiments report a comparison of selection rules rather than a rank-correlation coefficient. I cite the result as separate evidence that function-level freedom affects generalisation under unseen demands.

S5.4 Relation between K_ρ and neural freedom

$K_\rho(M)$ and $J_H(\phi)$ need not rank representations in the same order. The first depends on an evaluation distribution, diagnostic tests, and decision margins. The second depends on a frozen feature map, an empirical policy, an indexed bank of joint probe demands, and an affine-readout language. On the lifted diagnostic task of Section 4.5, K_ρ is a weighted freedom deficit because the lift removes cross-cell dependencies. The neural bank retains joint dependencies inside each bundle through the shared affine head, and each network has its own feature language.

A direct comparison therefore requires one external task template and one matched demand law for both diagnostics. Recurrent PPO study establishes that K_ρ can guide post-training representation selection on a lifted task. Remaining bridge question is whether joint completion adds predictive or selection value beyond K_ρ when one shared function class couples decisions across cells, and whether either quantity can be optimised during recurrent, convolutional, or transformer training.

S6 Symbol index

Table S8 lists the symbols used in the paper and where each is defined, so that any object appearing in a statement can be resolved without searching the text.

S7 Measurable reduction

The finite theorem has a direct measurable analogue. Let $(\mathcal{X}, \Sigma, \mu)$ be a probability space of history-test points $x = (h, T)$. Let

$$\zeta := (M(h), T)$$

be the observable representation-test variable. Assume $y : \mathcal{X} \rightarrow \{0, 1\}$ and $\rho : \mathcal{X} \rightarrow [0, 1]$ are measurable. Define

$$a(z) := \mathbb{E}[\rho \mathbf{1}\{y = 1\} \mid \zeta = z], \quad b(z) := \mathbb{E}[\rho \mathbf{1}\{y = 0\} \mid \zeta = z].$$

Any M -based policy is determined by a measurable map $q : \mathcal{Z} \rightarrow [0, 1]$. Its expected normalised regret is

$$R(q) := \mathbb{E}\left[\rho((1 - q(\zeta))\mathbf{1}\{y = 1\} + q(\zeta)\mathbf{1}\{y = 0\})\right].$$

Proposition S7.1 (Measurable reduction). *With the notation above,*

$$\inf_{q: \mathcal{Z} \rightarrow [0, 1] \text{ measurable}} R(q) = \mathbb{E}[\min\{a(\zeta), b(\zeta)\}].$$

An optimal measurable selector is given by

$$q^*(z) = \begin{cases} 1 & a(z) > b(z), \\ 0 & a(z) < b(z), \\ \text{any value in } [0, 1] & a(z) = b(z). \end{cases}$$

Table S8: Symbols, grouped by the reinforcement-learning side and the Stack-Theoretic side.

Symbol	Meaning
<i>Reinforcement-learning side</i>	
$M, M(h)$	Representation, and the representation state at history h
$\mathcal{M}$	Declared family of admissible representations (Corollary 3.15)
h, T, Z_T	History, diagnostic test, and its binary outcome
$p_T(h)$	$\Pr(Z_T = 1 \mid h)$, the true probability the test fires (Definition 3.1)
$x_i = (h_i, T_i)$	A support point of the evaluation distribution
μ_i, ρ_i	Evaluation mass and margin weight of x_i (Definitions 3.2 and 3.4)
$\delta_i = 1 - V_i^\pi / V_i^\star$	Normalised betting regret at x_i (Section 3.1)
q_C	Report probability of a policy on cell C
$\mathcal{C}_M, C$	The representation-test cells, and a generic cell
$K_\rho(M)$	Margin-weighted forgetting cost / minimum regret floor (Definition 3.8)
$A_C, B_C, A_C^{(k)}$	Margin-weighted label masses in cell C (Theorems 3.10 and 3.18)
$I_+, \mathcal{P}_+^\star$	Informative support and its coarsest pure diagnostic partition (Theorem 5.2)
<i>Stack-Theoretic side</i>	
$\Phi, p, \mathbf{v}$	States, a program $p \subseteq \Phi$, and a finite vocabulary (Definition 4.1)
$L_{\mathbf{v}}$	Embodied language of $\mathbf{v}$ (Definition 4.1)
π, ϑ, l	Statements (π and ϑ also serve as policies)
$\text{Ext}(\cdot), w(\cdot)$	Extension and freedom (Definition 4.2)
$\alpha, \omega, I_\alpha, O_\alpha, \Pi_\alpha$	Tasks, their inputs/outputs, and correct policies (Definition 4.3)
U, u_i	Unseen region, and an unseen output
$w_i, W(\cdot)$	Output weights, and prior-weighted freedom (Definition 4.7)
r_i, w_P, P	Relevance probabilities, prior-freedom, and the prior (Theorems S10.6 and S10.9)
$Z, W^\star(M)$	Total weight and maximum weighted freedom on the lifted task (Corollary 4.10)

Proof. By the tower property,

$$R(q) = \mathbb{E}[a(\zeta)(1 - q(\zeta)) + b(\zeta)q(\zeta)].$$

For each fixed z , the integrand is affine in $q(z)$. Its pointwise minimum over $q(z) \in [0, 1]$ is therefore $\min\{a(z), b(z)\}$. Choosing $q^\star$ pointwise attains that minimum almost surely. Integrability is immediate because $0 \leq a, b \leq \mathbb{E}[\rho \mid \zeta] \leq 1$. $\square$

This is the conditional-expectation form of the finite theorem. In the finite support case, ζ takes one value per representation-test cell, $a(C) = A_C/\mu(C)$ and $b(C) = B_C/\mu(C)$, so the outer expectation contributes the factor $\mu(C)$ and Theorem S7.1 reduces to $\sum_C \min\{A_C, B_C\}$. In measurable language, $\mathbb{E}[\min\{a(\zeta), b(\zeta)\}]$ is the weighted impurity that remains after quotienting by representation-test cell. It is the natural measurable version of $K_\rho(M)$.

S8 Contravariance and splintering on the diagnostic quotient

The bridge theorem contains a contravariance result. Let

$$I_+ := \{i : \mu_i \rho_i > 0\}, \quad \mathcal{T}_+ := \{T_i : i \in I_+\}.$$

Define the *coarsest pure diagnostic partition on the informative support*

$$\mathcal{P}_+^* := \{ \{i \in I_+ : T_i = t, y_i = b\} : t \in \mathcal{T}_+, b \in \{0, 1\} \} \setminus \{\emptyset\}.$$

So $\mathcal{P}_+^*$ groups positive-weight support points only by test and optimal label. Write $\mathcal{C} \preceq \mathcal{D}$ when every cell of partition $\mathcal{C}$ is contained in some cell of $\mathcal{D}$. For any diagnostic quotient $\mathcal{Q}$ of I_+ whose cells refine test value, write

$$K_\rho(\mathcal{Q}) := \sum_{C \in \mathcal{Q}} \min \left\{ \sum_{i \in C, y_i=1} \mu_i \rho_i, \sum_{i \in C, y_i=0} \mu_i \rho_i \right\}.$$

Proposition S8.1 (Zero-cost diagnostic quotients). *For a fixed support and label map y , the following hold.*

1. *For any diagnostic quotient $\mathcal{Q}$ of I_+ whose cells refine test value, $K_\rho(\mathcal{Q}) = 0$ iff $\mathcal{Q} \preceq \mathcal{P}_+^*$.*
2. *$\mathcal{P}_+^*$ is the unique coarsest zero-cost diagnostic quotient.*
3. *For an M -based history representation, $K_\rho(M) = 0$ iff $\mathcal{C}_M|_{I_+} \preceq \mathcal{P}_+^*$. If $\mathcal{P}_+^*$ is induced by some history representation, then minimal zero-cost history representations induce $\mathcal{P}_+^*$ up to recoding. In general they induce minimal feasible refinements of $\mathcal{P}_+^*$.*

Proof. If $K_\rho(M) = 0$, then every cell $C \in \mathcal{C}_M$ must satisfy $\min\{A_C, B_C\} = 0$ by Theorem 3.10. So on the positive-weight support I_+ each cell contains only one optimal label at its fixed test value, which is the statement that $\mathcal{C}_M|_{I_+} \preceq \mathcal{P}_+^*$. The converse is immediate from the definition of $K_\rho(M)$ because points outside I_+ attract zero deletion weight.

For any diagnostic quotient $\mathcal{Q}$, the same cellwise argument gives $K_\rho(\mathcal{Q}) = 0$ exactly when each quotient cell is contained in one cell of $\mathcal{P}_+^*$. Thus $\mathcal{P}_+^*$ is the unique coarsest zero-cost diagnostic quotient. For an M -based representation, apply this result to $\mathcal{Q} = \mathcal{C}_M|_{I_+}$. If $\mathcal{P}_+^*$ is M -realisable, coarsening any finer zero-cost induced quotient to $\mathcal{P}_+^*$ preserves zero cost, so minimality forces equality up to a recoding of representation states. If it is not M -realisable, equality is impossible and minimal zero-cost history representations are exactly the minimal induced refinements allowed by the constraint that one code $M(h)$ must be reused across all tests containing h . $\square$

Interpretation. Shared task pressure drives away weighted impurity. At zero impurity there is one coarsest diagnostic quotient. History encoders can require extra splits because their codes cannot change from one test to another.

Example S8.2 (Shared pressure versus splintering). Take three support points x_1, x_2, x_3 under one common test T . Subsystem A has optimal labels

$$y^A(x_1) = 1, \quad y^A(x_2) = 0, \quad y^A(x_3) = 0,$$

so its coarsest pure quotient is

$$\mathcal{P}_{A,+}^* = \{\{x_1\}, \{x_2, x_3\}\}.$$

Subsystem B has labels

$$y^B(x_1) = 0, \quad y^B(x_2) = 1, \quad y^B(x_3) = 0,$$

so its coarsest pure quotient is

$$\mathcal{P}_{B,+}^* = \{\{x_2\}, \{x_1, x_3\}\}.$$

These minimal zero-cost quotients do not coincide. If each subsystem separately minimises its own weighted forgetting cost, they converge to different recoding classes. If they are forced to share one common zero-cost quotient, they must instead move to the finer common refinement

$$\{\{x_1\}, \{x_2\}, \{x_3\}\}.$$

This is the abstract pattern behind the contrast used in the main text. Shared task pressure yields contravariant convergence to one common quotient. Split task pressure yields divergent locally sufficient quotients, which is the toy formal template for splintering and cancer-like coordination failure.

S9 Coordination cost on a shared diagnostic quotient

Combining Theorem 3.10 with the contravariance result (Appendix S8), I quantify the cost of sharing a representation across subsystems with incompatible diagnostic needs.

Proposition S9.1 (Coordination cost). *Let N subsystems each face a diagnostic family on the same support with the same single test T . Let $\mathcal{P}_j^*$ be the coarsest pure partition for subsystem j , with $|\mathcal{P}_j^*| = k_j$ cells. A shared representation that achieves zero regret for all N subsystems simultaneously requires at least $|\mathcal{P}_1^* \vee \dots \vee \mathcal{P}_N^*|$ cells, where $\vee$ denotes the common refinement. Its size is at most $\min\{|\mathcal{X}|, \prod_{j=1}^N k_j\}$, where $\mathcal{X}$ is the shared support. The product is attained when every tuple of component cells has nonempty intersection, which requires $|\mathcal{X}| \geq \prod_j k_j$.*

Proof. By Theorem 3.10, zero regret for subsystem j requires the shared partition to refine $\mathcal{P}_j^*$. The coarsest partition refining all N is the common refinement $\mathcal{P}_1^* \vee \dots \vee \mathcal{P}_N^*$. Each cell is determined by its class in every subsystem's partition, giving at most $\prod_j k_j$ possible tuples, and no partition can have more cells than the shared support. This gives the stated upper bound. The product is attained when every tuple of component cells has nonempty intersection. $\square$

Interpretation. Subsystems with incompatible diagnostic needs impose a combinatorial tax on shared representations. This connects to federated learning, where client drift [14] has the same structure, and to the biological literature on cancer-like coordination failure [4, 10, 11, 15].

Example S9.2 (Coordination cost in a two-subsystem scenario). Consider a domain with 6 support points under one common test T . Subsystem A requires the partition $\mathcal{P}_A^* = \{\{1, 2\}, \{3, 4, 5, 6\}\}$ ($k_A = 2$), while subsystem B requires $\mathcal{P}_B^* = \{\{1, 2, 3\}, \{4, 5, 6\}\}$ ($k_B = 2$). Each subsystem needs only 2 representation states, but a shared representation that achieves zero regret for both must have at least $|\mathcal{P}_A^* \vee \mathcal{P}_B^*| = |\{\{1, 2\}, \{3\}, \{4, 5, 6\}\}| = 3$ states. When the shared support contains every tuple of component cells, the shared representation requires $\prod_j k_j$ states, while subsystem j needs only k_j . This gap is the coordination tax.

S10 Imported Stack-Theoretic machinery

This appendix gives self-contained formal restatements of the Stack-Theoretic results that the main text imports. Each restatement uses the same notation and hypotheses as the master appendix [6]. Each result is stated and proved in full in this paper’s notation, so the cross-framework corollaries can be verified without consulting the source corpus. The original statements are in the cited sources.

This appendix lets a reader verify the cross-framework corollaries in Section 5.2 without reconstructing definitions from elsewhere in the corpus. A reader who accepts the imports can continue at Section S10.5.

S10.1 Minimal preliminaries

The following definitions are sufficient to state every imported result. They are taken verbatim from earlier work [4, 6].

Definition S10.1 (Environment, vocabulary, embodied language). An *environment* is a nonempty set Φ of mutually exclusive states. A *program* is a subset $p \subseteq \Phi$. A *vocabulary* is a finite set $\mathfrak{v} \subseteq 2^\Phi$ of programs. The *embodied language* induced by $\mathfrak{v}$ is the satisfiable fragment

$$L_{\mathfrak{v}} := \{ l \subseteq \mathfrak{v} \mid \bigcap_{p \in l} p \neq \emptyset \}.$$

Elements of $L_{\mathfrak{v}}$ are called *statements*. A statement is true at state ϕ when ϕ lies in the intersection of its programs.

Note that $L_{\mathfrak{v}}$ is the satisfiable fragment of $2^{\mathfrak{v}}$, not all of $2^{\mathfrak{v}}$. Statements correspond to physical configurations a body can be in, and most configurations are mutually exclusive. This restriction is what stops freedom from collapsing into a pure length proxy.

Definition S10.2 (Extension and freedom). For $l \in L_{\mathfrak{v}}$, the *extension* of l is

$$\text{Ext}(l) := \{ y \in L_{\mathfrak{v}} \mid l \subseteq y \}.$$

The *freedom* of l is $w(l) := |\text{Ext}(l)|$, the number of completions of l within $L_{\mathfrak{v}}$. A statement l_1 has more freedom than l_2 when $w(l_1) > w(l_2)$.

Freedom is intrinsic relative to a fixed embodied language $L_{\mathfrak{v}}$. It is not a function of external description length. The numerical freedom score $|\text{Ext}(l)|$ depends on the chosen embodied vocabulary. The encoding-invariant claim concerns the optimality of the freedom rule inside each finite vocabulary, not equality of freedom scores across vocabularies.

Definition S10.3 ($\mathfrak{v}$ -task, correct policy). For $X \subseteq L_{\mathfrak{v}}$, write

$$\text{Ext}(X) := \bigcup_{x \in X} \text{Ext}(x).$$

A $\mathfrak{v}$ -task is a pair $\alpha = \langle I_{\alpha}, O_{\alpha} \rangle$ of finite subsets, with $I_{\alpha} \subseteq L_{\mathfrak{v}}$ the inputs and $O_{\alpha} \subseteq \text{Ext}(I_{\alpha})$ the correct outputs. A *policy* is any statement $\pi \in L_{\mathfrak{v}}$. The policy is *correct* for α iff

$$\text{Ext}(I_{\alpha}) \cap \text{Ext}(\pi) = O_{\alpha}.$$

Write Π_{α} for the set of correct policies for α . Tasks α, ω satisfy $\alpha \sqsubset \omega$ when $I_{\alpha} \subseteq I_{\omega}$ and $O_{\alpha} \subseteq O_{\omega}$ (the parent extends the child).

S10.2 Optimality of freedom

The first imported block establishes that maximising freedom is sufficient and necessary for optimal generalisation under the maximally uninformative prior, and shows how the optimality survives weighted priors only conditionally.

Proposition S10.4 (Sufficiency of freedom under a maximally uninformative prior). *Let $\mathbf{v}$ be finite. Let α, ω be $\mathbf{v}$ -tasks with $\alpha \sqsubset \omega$. Write $U := L_{\mathbf{v}} \setminus \text{Ext}(I_{\alpha})$ for the unseen region. Assume the parent extends the child by selecting an additional set $S \subseteq U$ of correct outputs uniformly at random from 2^U , and that $O_{\omega} = O_{\alpha} \cup S$.*

Then, among $\pi \in \Pi_{\alpha}$, the probability that π is correct for ω is maximised by any policy with maximal $|\text{Ext}(\pi)|$.

Proof. Write $m := |U|$. By assumption S is uniform on 2^U , so $\Pr(S = S_0) = 2^{-m}$ for each $S_0 \subseteq U$. Since $\alpha \sqsubset \omega$ with $O_{\omega} = O_{\alpha} \cup S$ and $S \subseteq U$, a policy $\pi \in \Pi_{\alpha}$ is correct for ω exactly when its extension covers the newly required outputs, i.e. $S \subseteq \text{Ext}(\pi) \cap U$. Hence

$$\Pr(\pi \in \Pi_{\omega} \mid \alpha) = \Pr(S \subseteq \text{Ext}(\pi) \cap U).$$

For a fixed set $A := \text{Ext}(\pi) \cap U \subseteq U$, the admissible S contained in A are exactly the subsets of A , of which there are $2^{|A|}$, so $\Pr(S \subseteq A) = 2^{|A|}/2^m = 2^{|A|-m}$, strictly increasing in $|A|$. Every $\pi \in \Pi_{\alpha}$ satisfies $\text{Ext}(\pi) \cap \text{Ext}(I_{\alpha}) = O_{\alpha}$ by correctness on the seen region, so the seen part of the extension is common to all correct policies and any variation in $|\text{Ext}(\pi)|$ is carried entirely by $|\text{Ext}(\pi) \cap U|$. Maximising $|\text{Ext}(\pi)|$ over Π_{α} is therefore the same as maximising $|A|$, which maximises the survival probability. Any policy with maximal $|\text{Ext}(\pi)|$ thus maximises $\Pr(\pi \in \Pi_{\omega} \mid \alpha)$. $\square$

Proposition S10.5 (Necessity of freedom under the same prior). *Under the assumptions of Theorem S10.4, any $\pi \in \Pi_{\alpha}$ that maximises $\Pr(\pi \in \Pi_{\omega} \mid \alpha)$ must be $<_w$ -maximal in Π_{α} . Any non- $<_w$ -maximal correct policy is strictly suboptimal.*

Proof. Suppose $\pi \in \Pi_{\alpha}$ is not $<_w$ -maximal, so there is $\pi' \in \Pi_{\alpha}$ with $w(\pi') > w(\pi)$. As in Theorem S10.4, the seen parts of all correct policies coincide, so the difference in freedom is carried entirely by the unseen parts. Put $A := \text{Ext}(\pi) \cap U$ and $A' := \text{Ext}(\pi') \cap U$, so $|A| < |A'|$. Note that no inclusion between A and A' is needed, only the cardinality comparison. By the computation in Theorem S10.4,

$$\Pr(\pi \in \Pi_{\omega} \mid \alpha) = 2^{|A|-m} < 2^{|A'|-m} = \Pr(\pi' \in \Pi_{\omega} \mid \alpha),$$

so π is strictly suboptimal. Hence any maximiser of the generalisation probability is $<_w$ -maximal in Π_{α} . $\square$

Interpretation. These two propositions together justify the unconditional claim that the freedom-maximising correct policy is the generalisation-optimal choice under maximal ignorance about which unseen continuations matter. The argument is an enumerative argument with no prior structure other than uniformity.

Theorem S10.6 (Freedom maximisation beyond the uniform prior). *Let $\mathbf{v}$ be finite, $\alpha \sqsubset \omega$ as before, and $U = L_{\mathbf{v}} \setminus \text{Ext}(I_{\alpha})$. Let P be the conditional distribution of S given α , with no restriction to uniformity. Define the P -freedom of π by*

$$w_P(\pi) := P(S \subseteq \text{Ext}(\pi) \cap U) = \sum_{S \subseteq \text{Ext}(\pi) \cap U} P(S).$$

Then $\Pr(\pi \in \Pi_\omega \mid \alpha) = w_P(\pi)$, so $\pi^* \in \Pi_\alpha$ maximises generalisation probability iff it maximises $w_P(\pi)$ over Π_α .

If in addition P is exchangeable over U (meaning $P(S) = P(\sigma(S))$ for every permutation σ of U), then there exists a nondecreasing $f : \{0, 1, \dots, |U|\} \rightarrow [0, 1]$ such that $w_P(\pi) = f(|\text{Ext}(\pi) \cap U|)$, so any $<_w$ -maximal correct policy is also w_P -optimal under P .

Proof. As in Theorem S10.4, $\pi \in \Pi_\alpha$ is correct for ω iff $S \subseteq \text{Ext}(\pi) \cap U$, so

$$\Pr(\pi \in \Pi_\omega \mid \alpha) = P(S \subseteq \text{Ext}(\pi) \cap U) = \sum_{S \subseteq \text{Ext}(\pi) \cap U} P(S) = w_P(\pi),$$

which gives the first claim, and the equivalence of maximisers is immediate. For the second claim, write $A := \text{Ext}(\pi) \cap U$. If $A \subseteq A'$ then $\{S : S \subseteq A\} \subseteq \{S : S \subseteq A'\}$, so w_P is monotone under inclusion. If P is exchangeable then $P(S)$ depends only on $|S|$, say $P(S) = g(|S|)$. For any two sets of equal size there is a permutation σ of U sending one to the other, and w_P is invariant under σ because P is, so w_P depends only on $|A|$. Writing $f(k) := \sum_{j=0}^k \binom{k}{j} g(j)$ (the $\binom{k}{j}$ subsets of size j inside a set of size k), this gives $w_P(\pi) = f(|A|)$, and monotonicity under inclusion forces f nondecreasing. Hence any $<_w$ -maximal correct policy maximises $|A|$ and therefore maximises w_P . $\square$

Remark S10.7 (Weighted freedom is not a free upgrade). Theorem S10.6 shows that freedom extends beyond uniform priors only as far as the prior remains exchangeable. Once you adopt a non-exchangeable independent prior $r_i \in [0, 1)$ over individual unseen outputs, the Bayes-optimal rule is weighted freedom with weights $w_i = -\log(1 - r_i)$ [6, Proposition on weighted prior]. That weighting matches the prior you adopted. It is not generalisation-optimal under any other prior. In particular, the canonical choice $r_i = 1 - e^{-\mu_i \rho_i}$ used in Corollary 5.4 is the prior under which the regret floor $K_\rho(M)$ has the exponential meaning $e^{-K_\rho(M)}$. Plain freedom remains generalisation-optimal under uniform ignorance.

Freedom is invariant under function-preserving parameter reparameterisations once the embodied language and measure are fixed [7]. A change of embodied vocabulary changes the completion relation itself, so $|\text{Ext}(\pi)|$ can change. The proof of Theorem S10.4 yields, inside any finite vocabulary $\mathfrak{v}$,

$$\Pr(\pi \in \Pi_\omega \mid \alpha) = \frac{2^{|\text{Ext}(\pi) \cap U|}}{2^{|U|}}.$$

This is monotone in $|\text{Ext}(\pi) \cap U|$, so the freedom-maximising correct policy is generalisation-optimal under maximal ignorance in every embodied language.

Alternative rules do not have that property. Description-length minimisation can pick a non-freedom-maximising policy in some vocabularies and is therefore not generalisation-optimal there [6, 8, Propositions on simplicity sub-optimality and subjectivity of description length]. Weighted freedom is optimal under a specific prior P , but the prior is pinned to the unseen outputs u_i of a particular language. Re-encode and the unseen region changes, the weights have to be re-specified for the new u_i , and whether the new weighted rule is optimal in the new language depends on whether the new prior matches the new environment. Plain freedom's optimality survives encoding changes. Alternatives' effectiveness does not.

So adopting a weighting buys specialisation at the cost of two things. It forfeits unconditional generalisation optimality. And it forfeits the encoding-invariance of that optimality, because the prior must be re-specified across encodings and need not remain correct. This is the same observation as Remark 4.13 in the main text. It is repeated here so that readers consulting only this appendix do not over-read the optimality of weighted freedom.

S10.3 Weighted freedom and the bridge weights

This is the result the main text imports as Proposition 4.9. It is restated here in self-contained form so the bridge corollary (Corollary 4.10) can be checked against the master appendix directly.

Definition S10.8 (Prior-weighted freedom). Let $U = \{u_1, \dots, u_n\}$ be a finite unseen region with weights $w_1, \dots, w_n \geq 0$. For any policy $\vartheta \in L_{\mathbf{v}}$, the *prior-weighted freedom* of ϑ on U is

$$W(\vartheta) := \sum_{u_i \in \text{Ext}(\vartheta) \cap U} w_i.$$

Proposition S10.9 (Independent nonuniform priors give weighted freedom). *Assume each unseen output $u_i \in U$ becomes relevant independently with probability $0 \leq r_i < 1$. Then for any correct child policy ϑ ,*

$$\log \Pr(\vartheta \text{ generalises}) = \sum_{u_i \notin \text{Ext}(\vartheta) \cap U} \log(1 - r_i).$$

Define

$$w_i := -\log(1 - r_i).$$

Maximising generalisation probability under this prior is equivalent to maximising

$$\sum_{u_i \in \text{Ext}(\vartheta) \cap U} w_i,$$

which is prior-weighted freedom.

Proof. A correct child policy ϑ fails to generalise exactly when some omitted unseen output becomes relevant. Equivalently, ϑ generalises iff every $u_i \notin \text{Ext}(\vartheta) \cap U$ is irrelevant. Under the independent prior these events are independent with probabilities $1 - r_i$, so

$$\Pr(\vartheta \text{ generalises}) = \prod_{u_i \notin \text{Ext}(\vartheta) \cap U} (1 - r_i), \quad \log \Pr(\vartheta \text{ generalises}) = \sum_{u_i \notin \text{Ext}(\vartheta) \cap U} \log(1 - r_i).$$

The total $\sum_{u_i \in U} \log(1 - r_i)$ does not depend on ϑ , so maximising the left-hand side is the same as maximising $\sum_{u_i \in \text{Ext}(\vartheta) \cap U} (-\log(1 - r_i))$, which is the prior-weighted freedom $W(\vartheta)$ with weights $w_i := -\log(1 - r_i)$. $\square$

Interpretation. This is the weighted version of Theorem S10.4. Uniform priors recover ordinary freedom. Biased priors recover weighted freedom. As Theorem S10.7 stresses, the optimality is conditional on the prior matching the actual environment.

S10.4 The EGRL freedom pair proxy

This subsection records the EGRL construction the main text refers to, namely the enactive general reinforcement learning machinery in the thesis [6], together with the optimality theorem for the freedom pair proxy. The original statements are in earlier work [3, 5, 6, 8].

Definition S10.10 (Enactive General Reinforcement Learning, EGRL). Fix a vocabulary $\mathbf{v}$, a reward predicate $r : L_{\mathbf{v}} \rightarrow \{0, 1\}$, and a finite horizon $len \in \mathbb{N}$ with $len > 0$. The reward predicate induces a $\mathbf{v}$ -task $\mu = \langle I_{\mu}, O_{\mu} \rangle$ by

$$O_{\mu} := \{l \in L_{\mathbf{v}} \mid r(l) = 1\}, \quad I_{\mu} := \{i \in L_{\mathbf{v}} \mid \exists o \in O_{\mu} (i \subseteq o)\}.$$

Assume $\Pi_\mu \neq \emptyset$. Interaction up to time t is modelled by a history-task $\mathfrak{h}_{<t} \in \Gamma_{\mathfrak{v}}$. Apply r to past outputs to split into positive and negative subhistories

$$O_{\mathfrak{h}_{<t}}^1 := \{o \in O_{\mathfrak{h}_{<t}} \mid r(o) = 1\}, \quad O_{\mathfrak{h}_{<t}}^0 := O_{\mathfrak{h}_{<t}} \setminus O_{\mathfrak{h}_{<t}}^1,$$

and lift inputs accordingly. The unseen region is $U_{\mathfrak{h}_{<t}} := L_{\mathfrak{v}} \setminus \text{Ext}(I_{\mathfrak{h}_{<t}})$.

A *pair proxy* is a binary relation $<^{\text{pp}}$ on the admissible set

$$\text{pp} := \{ (m, n) \in \Pi_{\mathfrak{h}_{<t}}^1 \times \Pi_{\mathfrak{h}_{<t}}^0 \mid \text{Ext}(m) \cap \text{Ext}(n) = \emptyset \}.$$

Learning means choosing a pair maximal under $<^{\text{pp}}$. *Inference* given current input $i_t \in I_\mu$ means choosing $o_t \in \text{Ext}(i_t) \cap \text{Ext}(\pi)$ where π is the positive component of the chosen pair.

The *freedom pair proxy*, called the weakness pair proxy in the earlier EGRL papers, ranks pairs by joint extension on the unseen region, via

$$(j, k) <_w^{\text{pp}} (\pi, n) \quad \text{iff} \quad |(\text{Ext}(j) \cup \text{Ext}(k)) \cap U_{\mathfrak{h}_{<t}}| < |(\text{Ext}(\pi) \cup \text{Ext}(n)) \cap U_{\mathfrak{h}_{<t}}|.$$

Interpretation. An EGRL system maintains a positive policy and a disjoint negative policy. The positive policy prescribes what to do, the negative policy proscribes what to avoid. A pair proxy ranks admissible disjoint pairs. The freedom pair proxy ranks them by how much of the unseen future stays compatible with at least one of the two components.

Proposition S10.11 (Freedom pair proxy optimality). *Let $\mathfrak{h}_{<t}$ be a history-task with induced subhistories $\mathfrak{h}_{<t}^1, \mathfrak{h}_{<t}^0$. Let pp be the admissible disjoint pair set, $E_{\pi, n} := \text{Ext}(\pi) \cup \text{Ext}(n)$ the effective completion set, and $U_{\mathfrak{h}_{<t}}$ as above. Under the maximally uninformative extension model of Theorem S10.4 applied to $E_{\pi, n}$, the probability that an admissible pair generalises to an unseen parent is maximised by choosing $(\pi, n) \in \text{pp}$ with maximal $|E_{\pi, n} \cap U_{\mathfrak{h}_{<t}}|$.*

If in addition $E_{\pi, n} \cap \text{Ext}(I_{\mathfrak{h}_{<t}})$ is the same set for every admissible pair, then maximising $|E_{\pi, n} \cap U_{\mathfrak{h}_{<t}}|$ is equivalent to maximising $|E_{\pi, n}|$.

Proof. The argument is the enumerative argument of Theorem S10.4 with the single extension $\text{Ext}(\pi)$ replaced by the effective completion set $E_{\pi, n} := \text{Ext}(\pi) \cup \text{Ext}(n)$. Under the maximally uninformative model the parent selects an additional set $S \subseteq U_{\mathfrak{h}_{<t}}$ uniformly at random. An admissible pair (π, n) remains consistent with that parent exactly when every selected unseen output is allowed by at least one component, i.e. when $S \subseteq E_{\pi, n} \cap U_{\mathfrak{h}_{<t}}$. Writing $A := E_{\pi, n} \cap U_{\mathfrak{h}_{<t}}$ and $m := |U_{\mathfrak{h}_{<t}}|$, the computation of Theorem S10.4 gives

$$\Pr((\pi, n) \text{ generalises}) = \Pr(S \subseteq A) = 2^{|A|-m},$$

strictly increasing in $|A| = |E_{\pi, n} \cap U_{\mathfrak{h}_{<t}}|$. Hence the generalisation probability is maximised by the admissible pair with the largest unseen effective completion set, which is exactly the ranking computed by the freedom pair proxy. If moreover $E_{\pi, n} \cap \text{Ext}(I_{\mathfrak{h}_{<t}})$ is common to all admissible pairs, then $|E_{\pi, n}| = |E_{\pi, n} \cap \text{Ext}(I_{\mathfrak{h}_{<t}})| + |E_{\pi, n} \cap U_{\mathfrak{h}_{<t}}|$ varies across pairs only through the second term, so maximising $|E_{\pi, n} \cap U_{\mathfrak{h}_{<t}}|$ is equivalent to maximising $|E_{\pi, n}|$. $\square$

S10.5 Genealogy summary

The above subsections give formal statements for the imports the main text uses. The following is a one-paragraph genealogy locating those imports in the broader corpus and clarifying what is not imported.

The Sufficiency, Necessity, and beyond-uniform-prior results (Theorems S10.4 to S10.6) appeared in 2023 and were extended in the thesis and master appendix [6, 8]. The EGRL history-task construction (Theorem S10.10), its selective-memory clauses, and the freedom pair proxy optimality result (Theorem S10.11) appeared between 2024 and 2025 [3, 5, 6, 8]. The intervention-sensitive causal line (emergent causality, w -maximised causal identities, the do-operator-as-conditioning result, and the higher-order self convergence) is also in the corpus but is not imported by this paper, so it is not restated here [6, 8].

Nayebi’s structural selection theorems for predictive state, memory, modularity, regime tracking, and recoding match [17] are distinctive contributions of his programme without direct Stack-Theoretic antecedents. While the Richens–Everitt robust-causal-model result under distributional shift [18] may be seen as translating the Stack-Theoretic causal identity and optimal learning results [2] to a reinforcement learning setting, no formal bridge was presented.

References

- [1] Michael Timothy Bennett. The optimal choice of hypothesis is the weakest, not the shortest. In *16th International Conference on Artificial General Intelligence*, Lecture Notes in Computer Science, pages 42–51. Springer, 2023. doi: 10.1007/978-3-031-33469-6_5. URL https://doi.org/10.1007/978-3-031-33469-6_5.
- [2] Michael Timothy Bennett. Emergent causality and the foundation of consciousness. In *16th International Conference on Artificial General Intelligence*, Lecture Notes in Computer Science, pages 52–61. Springer, 2023. doi: 10.1007/978-3-031-33469-6_6. URL https://doi.org/10.1007/978-3-031-33469-6_6.
- [3] Michael Timothy Bennett. Computational dualism and objective superintelligence. In *17th International Conference on Artificial General Intelligence*, Lecture Notes in Computer Science. Springer, 2024. doi: 10.1007/978-3-031-65572-2_3. URL https://doi.org/10.1007/978-3-031-65572-2_3.
- [4] Michael Timothy Bennett. Are biological systems more intelligent than artificial intelligence?, 2024. URL <https://arxiv.org/abs/2405.02325>. In press, Philosophical Transactions of the Royal Society B: Biological Sciences. Special issue on Hybrid agencies: crossing borders between biological and artificial worlds.
- [5] Michael Timothy Bennett. Optimal policy is weakest policy. In *Artificial General Intelligence*, volume 16057 of *Lecture Notes in Computer Science*, pages 43–56. Springer, 2025. doi: 10.1007/978-3-032-00686-8_5. URL https://doi.org/10.1007/978-3-032-00686-8_5.
- [6] Michael Timothy Bennett. *How To Build Conscious Machines*. PhD thesis, The Australian National University, 2025. URL <https://hdl.handle.net/1885/733782452>.
- [7] Michael Timothy Bennett. Are flat minima an illusion?, 2026. URL <https://arxiv.org/abs/2605.05209>.
- [8] Michael Timothy Bennett. *How to Build Conscious Machines: Deriving Intelligence, Life and Mind from Time*. Palgrave Macmillan, 2027. Forthcoming.
- [9] Byron Boots, Sajid M. Siddiqi, and Geoffrey J. Gordon. Closing the learning-planning loop with predictive state representations. *International Journal of Robotics Research*, 30(7):954–966, 2011.

- [10] P C W Davies and C H Lineweaver. Cancer tumors as metazoa 1.0: tapping genes of ancient ancestors. *Physical Biology*, 8(1), February 2011. doi: 10.1088/1478-3975/8/1/015001. URL <https://doi.org/10.1088/1478-3975/8/1/015001>.
- [11] Chris Fields and Michael Levin. Scale-free biology: Integrating evolutionary and developmental thinking. *BioEssays*, 42, 06 2020. doi: 10.1002/bies.201900228. URL <https://doi.org/10.1002/bies.201900228>.
- [12] David Ha and Jürgen Schmidhuber. Recurrent world models facilitate policy evolution. In *Advances in Neural Information Processing Systems 31*, pages 2451–2463. Curran Associates, Inc., 2018. URL <https://papers.nips.cc/paper/7512-recurrent-world-models-facilitate-policy-evolution>. <https://worldmodels.github.io>.
- [13] Danijar Hafner, Timothy Lillicrap, Jimmy Ba, and Mohammad Norouzi. Dream to control: Learning behaviors by latent imagination. In *International Conference on Learning Representations*, 2020.
- [14] Sai Praneeth Karimireddy, Satyen Kale, Mehryar Mohri, Sashank Reddi, Sebastian Stich, and Ananda Theertha Suresh. SCAFFOLD: Stochastic controlled averaging for federated learning. In *Proceedings of the 37th International Conference on Machine Learning*, pages 5132–5143, 2020.
- [15] Michael Levin. The computational boundary of a “self”: Developmental bioelectricity drives multicellularity and scale-free cognition. *Frontiers in Psychology*, 10:2688, 2019.
- [16] Michael L. Littman, Richard S. Sutton, and Satinder Singh. Predictive representations of state. In *Advances in Neural Information Processing Systems*, volume 14, pages 1555–1561, 2001.
- [17] Aran Nayebi. What capable agents must know: Selection theorems for robust decision-making under uncertainty. In *Proceedings of the 42nd Conference on Uncertainty in Artificial Intelligence*, volume 337 of *Proceedings of Machine Learning Research*, pages 4773–4795. PMLR, 2026. URL <https://proceedings.mlr.press/v337/nayebi26a.html>.
- [18] Jonathan Richens and Tom Everitt. Robust agents learn causal world models. In *The Twelfth International Conference on Learning Representations*, 2024. doi: 10.48550/arxiv.2402.10877. URL <https://arxiv.org/abs/2402.10877>.
- [19] Jonathan Richens, Tom Everitt, and David Abel. General agents need world models. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 51659–51687. PMLR, 2025. URL <https://proceedings.mlr.press/v267/richens25a.html>.
- [20] Julian Schrittwieser, Ioannis Antonoglou, Thomas Hubert, Karen Simonyan, Laurent Sifre, Simon Schmitt, Arthur Guez, Edward Lockhart, Demis Hassabis, Thore Graepel, Timothy Lillicrap, and David Silver. Mastering Atari, Go, chess and shogi by planning with a learned model. *Nature*, 588(7839):604–609, 2020.
- [21] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017. doi: 10.48550/arXiv.1707.06347. URL <https://arxiv.org/abs/1707.06347>.

- [22] Satinder Singh, Michael R. James, and Matthew R. Rudary. Predictive state representations: A new theory for modeling dynamical systems. In *Proceedings of the 20th Conference on Uncertainty in Artificial Intelligence*, pages 512–519. AUAI Press, 2004.
- [23] Jayakumar Subramanian, Arunesh Sinha, Razieh Seraj, and Aditya Mahajan. Approximate information state for approximate planning and reinforcement learning in partially observed systems. *Journal of Machine Learning Research*, 23(12):1–83, 2022.